Alexandra Chang

STAT525 HOMEWORK#5

1. Use the **patient satisfaction** data described in KNNL Problem 6.15.

   a Compute the pairwise correlations between the X's and between each X and Y. Which X variable appears to be the best individual predictor?

   b Run the linear regression with age, severity of illness and anxiety level as the explanatory variables and satisfaction as the response variable. Summarize the regression results.

   c Plot the residuals versus the predicted satisfaction and each of the explanatory variables. Are there any unusual patterns?

   d Examine the assumption of normality for the residuals using a qqplot or histogram. State your conclusions.

   e Predict the satisfaction for a 55 year old patient with illness severity 50 and anxiety level 2.8. Provide a 95% prediction interval with your prediction.

2. KNNL Problem 7.5

3. KNNL Problem 7.6 (Please try to only use SAS statement `model y = x1 x2 x3/ss1;`, and derive the test based on it. Then justify your answer with `test x2 x3;`)

4. Derive the equation (7.56) on page 281 (HINT: Recall that b_1 can be obtained by regressing the residuals of $Y|X_2$ vs the residuals of $X_1|X_2$) (STAT Majors)

5. KNNL Problem 8.6

6. KNNL Problem 8.15

cli - Confidence Limits for Individual (Prediction Interval)
clm - Confidence Limits for Mean (CI for Mean Response)
clb - Confidence Limits for Regression Coefficients

1. Use the **patient satisfaction** data described in KNNL Problem 6.15.

   a Compute the pairwise correlations between the X's and between each X and Y. Which X variable appears to be the best individual predictor?

```
/*************************************************************************
 * Compute pairwise correlations between X's and between each X and Y
 ************************************************************************

proc corr data=a1 noprob;
    var x1 x2 x3 y;
run;
```

The CORR Procedure

4 Variables:	x1 x2 x3 y

Simple Statistics

Variable	N	Mean	Std Dev	Sum	Minimum	Maximum
x1	46	61.56522	17.23646	2832	26.00000	92.00000
x2	46	38.39130	8.91809	1766	22.00000	55.00000
x3	46	50.43478	4.31356	2320	41.00000	62.00000
y	46	2.28696	0.29934	105.20000	1.80000	2.90000

Pearson Correlation Coefficients, N = 46

	x1	x2	x3	y
x1	1.00000	-0.78676	-0.60294	-0.64459
x2	-0.78676	1.00000	0.56795	0.56968
x3	-0.60294	0.56795	1.00000	0.67053
y	-0.64459	0.56968	0.67053	1.00000

- The strongest correlation with Y is X_3 (Anxiety Level), $r = 0.671$
- This suggests X_3 (Anxiety Level) is the best predictor for patient satisfaction.
- However, correlations among the X's show moderate collinearity.
  Ex: X_2 v.s X_1, $r = -0.79$

   b Run the linear regression with age, severity of illness and anxiety level as the explanatory variables and satisfaction as the response variable. Summarize the regression results.

```
/*****************************************
 * Multiple Regression: Y = X1 + X2 + X3
 *****************************************

proc reg data=a1;
    model y = x1 x2 x3;
run;
```

The REG Procedure
Model: MODEL1
Dependent Variable: y

Number of Observations Read	46
Number of Observations Used	46

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	2.18362	0.72787	16.54	<.0001
Error	42	1.84856	0.04401		
Corrected Total	45	4.03217			

Root MSE	0.20979	R-Square	0.5415
Dependent Mean	2.28696	Adj R-Sq	0.5088
Coeff Var	9.17348		

Parameter Estimates

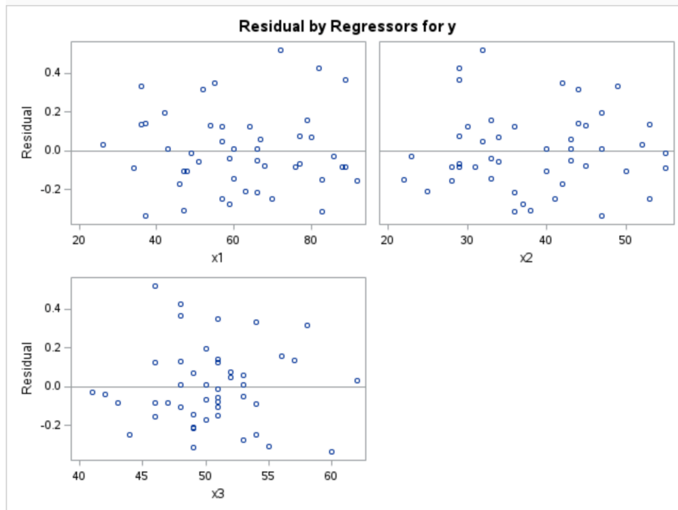
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	1.05325	0.61379	1.72	0.0935
x1	1	-0.00586	0.00309	-1.90	0.0647
x2	1	0.00193	0.00579	0.33	0.7407
x3	1	0.03015	0.00926	3.26	0.0022

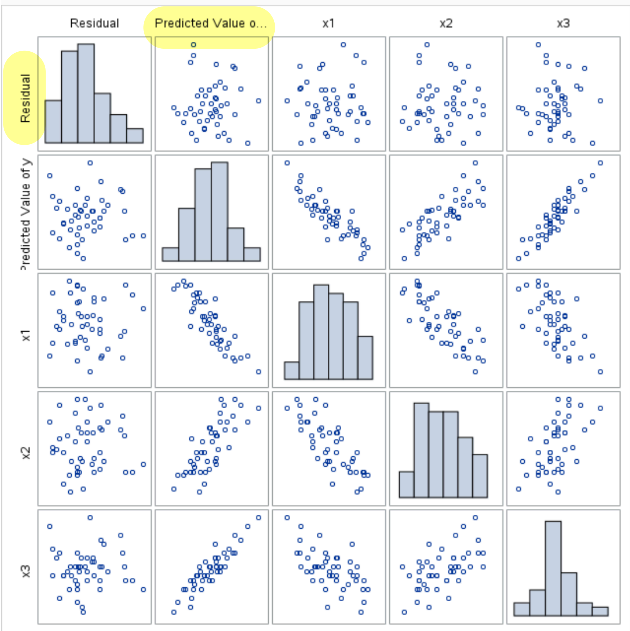
$$\hat{Y} = 1.053 - 0.0059X_1 + 0.0019X_2 + 0.0302X_3$$

- The overall model is significant ($F = 16.54$, $p < 0.0001$), indicating that age, severity, and anxiety jointly explain the variation in patient satisfaction.
- $R^2 = 0.54$: 54% variation in patient satisfaction is explained by age, severity, and anxiety.
- Age (X_1) has a negative but marginally significant effect ($p = 0.065$), indicating that older patients tend to report slightly lower patient satisfaction.
- Severity of Illness (X_2) not significant ($p = 0.74$).
- Anxiety Level (X_3) is significant, indicating that higher anxiety tend to report higher patient satisfaction.

c Plot the residuals versus the predicted satisfaction and each of the explanatory variables. Are there any unusual patterns?

```
/********************************************************
 * Residuals versus predicted and explanatory variables
 ********************************************************
proc sgscatter data=out1;
    matrix resid pred x1 x2 x3 / diagonal=(histogram);
run;
```

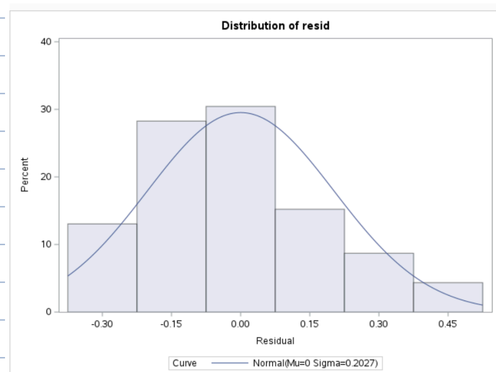




- Residual Plots seems to all be scattering around zero across $\hat{Y}$ and X_1, X_2, X_3.
- No obvious curvature or funnel pattern, so linearity and constant variance may be satisfied.
- No major outliers observed.

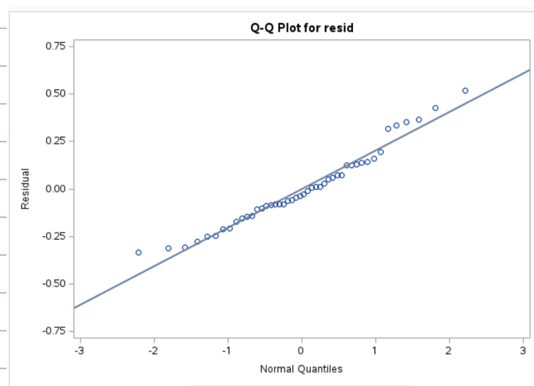
d Examine the assumption of normality for the residuals using a qqplot or histogram. State your conclusions.

```
/********************************************************
 * Normality Check for Residuals
 ********************************************************
proc univariate data=out1 normal;
    var resid;
    histogram resid / normal;
    qqplot resid / normal(mu=est sigma=est);
run;
```



Residuals appear to follow normal distribution, no serious violations of the model's normality assumption.

- Histogram of ei approx. bell-shaped and centered near 0, matching fitted normal curve well.
- Q-Q Plot points lie approx. along the diagonal referenced line with only minor deviations at tails.



e Predict the satisfaction for a 55 year old patient with illness severity 50 and anxiety level 2.8. Provide a 95% prediction interval with your prediction.

```
/*************************************************************************
 * Prediction and 95% Prediction Interval
 *************************************************************************/
/* Step 1: New observation */
data new;
    input x1 x2 x3;
    datalines;
55 50 2.8
;
run;

/* Step 2: Combine original data + new observation */
data combined;
    set a1 new;
run;

/* Step 3: Fit model and compute prediction + 95% PI */
proc reg data=combined;
    model y = x1 x2 x3 / cli;
    id x1 x2 x3;
    output out=pred p=predicted lcl=lcl95 ucl=ucl95;
run;
quit;

/* Step 4: Print only the new case */
proc print data=pred;
    where missing(y);  /* this filters to the new observation (since y is missing) */
    var x1 x2 x3 predicted lcl95 ucl95;
    title "Predicted Satisfaction and 95% Prediction Interval for New Patient";
run;
```

Predicted Satisfaction and 95% Prediction Interval for New Patient

Obs	x1	x2	x3	predicted	lcl95	ucl95
47	55	50	2.8	0.91173	-0.11576	1.93923

• Patient satisfaction level when 55 year old with illness severity 50 and anxiety 2.8 = 0.912, with PI: (-0.11576, 1.93923)

2. KNNL Problem 7.5

(Sequential Type I)
order: X_2, X_1, X_3

***7.5.** Refer to **Patient satisfaction** Problem 6.15.

a. Obtain the analysis of variance table that decomposes the regression sum of squares into extra sums of squares associated with X_2; with X_1, given X_2; and with X_3, given X_2 and X_1.

```
/*************************************************************************
 * 2a: Obtain extra SS for X2, X1|X2, and X3|X2,X1
 *************************************************************************
/* Order 1: X2 entered first, then X1, then X3 */
title "Decomposition of Regression SS into Extra Sums of Squares";
proc reg data=a1;
    model y = x2 x1 x3 / ss1 ss2;
run;
```

$SS = SS(X_2) + SS(X_1|X_2) + SS(X_3|X_1,X_2)$

• $SS(X_2) = 1.30857$

• $SS(X_1|X_2) = 0.40818$

• $SS(X_3|X_1,X_2) = 0.46686$

Add up to: 2.18362

Decomposition of Regression SS into Extra Sums of Squares

The REG Procedure
Model: MODEL1
Dependent Variable: y

Number of Observations Read	46
Number of Observations Used	46

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	2.18362	0.72787	16.54	<.0001
Error	42	1.84856	0.04401		
Corrected Total	45	4.03217			

Root MSE	0.20979	R-Square	0.5415
Dependent Mean	2.28696	Adj R-Sq	0.5088
Coeff Var	9.17348		

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Type I SS	Type II SS
Intercept	1	1.05325	0.61379	1.72	0.0935	240.58783	0.12960
x2	1	0.00193	0.00579	0.33	0.7407	1.30857	0.00489
x1	1	-0.00586	0.00309	-1.90	0.0647	0.40818	0.15844
x3	1	0.03015	0.00926	3.26	0.0022	0.46686	0.46686

b. Test whether X_3 can be dropped from the regression model given that X_1 and X_2 are retained. Use the F^* test statistic and level of significance .025. State the alternatives, decision rule, and conclusion. What is the P-value of the test?

$H_0: \beta_3 = 0$ (X_3 doesn't contribute additional explanatory power)
$H_A: \beta_3 \neq 0$ (X_3 contribute additional explanatory power after accounting for $X_1 + X_2$)

GLM in terms of Extra SS:

$$\boxed{F^* = \frac{SS(X_3|X_1,X_2)/(3-2)}{MSE}}$$

$$F^* = \frac{0.46686/1}{0.04401} = 10.61 \sim F^*_{1,42} > 5.17 \sim F_{0.025,1,42}$$

p-value = 0.0022 < 0.025

one-way

Reject Ho. Therefore, X_3 contribute additional explanatory power, even after accounting for $X_1 + X_2$.

3. KNNL Problem 7.6 (Please try to only use SAS statement `model y = x1 x2 x3/ss1;`, and derive the test based on it. Then justify your answer with `test x2 x3;`)

*7.6. Refer to **Patient satisfaction** Problem 6.15. Test whether both X_2 and X_3 can be dropped from the regression model given that X_1 is retained. Use $\alpha = .025$. State the alternatives, decision rule, and conclusion. What is the P-value of the test?

```
/************************************
 * 3: Test contribution of X2 and X3
 ************************************
proc reg data=a1;
    model y = x1 x2 x3 / ss1;
    test x2, x3;
run;
```

The REG Procedure
Model: MODEL1
Dependent Variable: y

Number of Observations Read	46
Number of Observations Used	46

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	2.18362	0.72787	16.54	<.0001
Error	42	1.84856	0.04401		
Corrected Total	45	4.03217			

Root MSE	0.20979	R-Square	0.5415
Dependent Mean	2.28696	Adj R-Sq	0.5088
Coeff Var	9.17348		

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Type I SS
Intercept	1	1.05325	0.61379	1.72	0.0935	240.58783
x1	1	-0.00586	0.00309	-1.90	0.0647	1.67536
x2	1	0.00193	0.00579	0.33	0.7407	0.04139
x3	1	0.03015	0.00926	3.26	0.0022	0.46686

$H_0: \beta_2 = \beta_3 = 0$
H_A: At least one β_2 or $\beta_3 \neq 0$

GLM in terms of Extra SS:

$$\boxed{F^* = \frac{SS(X_1,X_2|X_3)/(3-1)}{MSE}}$$

$$F^* = \frac{0.50826/2}{0.04401} = 5.77 \sim F^*_{2,42} > 3.23 \sim F_{0.025,2,42}$$

p-value = 0.0067 < 0.025

one-way

Reject Ho. Therefore, At least one X_2 or X_3 contribute additional explanatory power, even after accounting for X_1.

$SS(X_1,X_2|X_3) = 0.04139 + 0.46686 = 0.50826$

4. Derive the equation (7.56) on page 281 (HINT: Recall that b_1 can be obtained by regressing the residuals of $Y|X_2$ vs the residuals of $X_1|X_2$)

(Skipped)

To show that the regression coefficient of X_1 is unchanged when X_2 is added to the regression model in the case where X_1 and X_2 are uncorrelated, consider the following algebraic expression for b_1 in the first-order multiple regression model with two predictor variables:

$$b_1 = \frac{\dfrac{\sum(X_{i1}-\bar{X}_1)(Y_i-\bar{Y})}{\sum(X_{i1}-\bar{X}_1)^2} - \left[\dfrac{\sum(Y_i-\bar{Y})^2}{\sum(X_{i1}-\bar{X}_1)^2}\right]^{1/2} r_{Y2}r_{12}}{1-r_{12}^2} \tag{7.56}$$

Step-by-Step Derivation of Equation (7.56)

Step 1: Start with a multiple regression model

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \varepsilon_i$$

We want to derive an expression for β_1 (denoted as b_1 in sample terms).

Step 2: Apply the FWL Theorem

According to FWL, you can estimate b_1 by:

$$b_1 = \frac{\sum(X_{1i}^* - \bar{X}_1^*)(Y_i^* - \bar{Y}^*)}{\sum(X_{1i}^* - \bar{X}_1^*)^2}$$

Where:

- X_1^* = residuals from regressing X_1 on X_2
- Y^* = residuals from regressing Y on X_2

Step 3: Use residual properties

These residuals can be standardized, and the sample covariance becomes a function of correlation coefficients.

Let:

- r_{12}: correlation between X_1 and X_2
- r_{Y2}: correlation between Y and X_2

Then:

$$b_1 = \frac{S_{X_1Y} - r_{Y2}r_{12}S_Y}{S_{X_1}(1-r_{12}^2)}$$

Rewriting in terms of **sums of squares**:

Step 4: Equation (7.56)

$$b_1 = \frac{\sum(X_{1i}-\bar{X}_1)(Y_i-\bar{Y}) - \left[\frac{\sum(Y_i-\bar{Y})^2}{\sum(X_{1i}-\bar{X}_1)^2}\right]^{1/2} \cdot r_{Y2}r_{12}}{1-r_{12}^2}$$

This matches **equation (7.56)** exactly.

Interpretation:

If $r_{12} = 0$, then the second term becomes 0, and:

$$b_1 = \frac{\sum(X_{1i}-\bar{X}_1)(Y_i-\bar{Y})}{\sum(X_{1i}-\bar{X}_1)^2}$$

which is the **simple linear regression coefficient** for regressing Y on X_1 alone.
Hence, **when X_1 and X_2 are uncorrelated, the coefficient of X_1 does not change** whether or not X_2 is included.

5. KNNL Problem 8.6

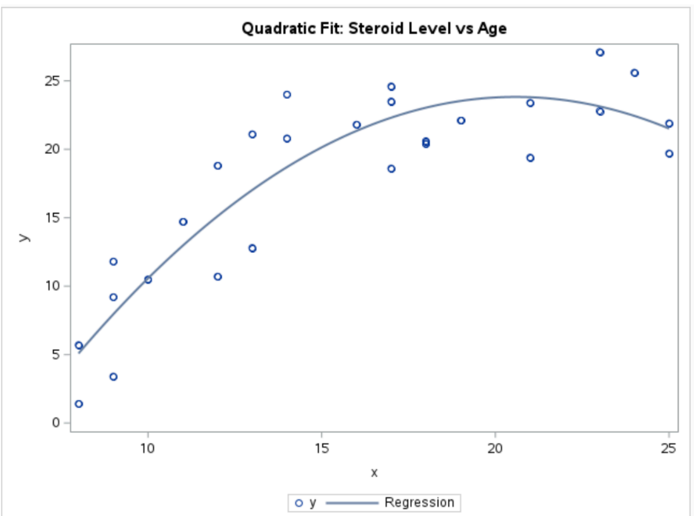
8.6. Steroid level. An endocrinologist was interested in exploring the relationship between the level of a steroid (Y) and age (X) in healthy female subjects whose ages ranged from 8 to 25 years. She collected a sample of 27 healthy females in this age range. The data are given below:

i:	1	2	3	...	25	26	27
X_i:	23	19	25	...	13	14	18
Y_i:	27.1	22.1	21.9	...	12.8	20.8	20.6

a. Fit regression model (8.2). Plot the fitted regression function and the data. Does the quadratic regression function appear to be a good fit here? Find R^2.

```
proc reg data=steroid;
    model y = x x2 / clb;
    plot y*x;
    title "Quadratic Regression Model: Steroid Level vs Age";
run;

proc sgplot data=steroid;
    scatter x=x y=y;
    reg x=x y=y / degree=2;
    title "Quadratic Fit: Steroid Level vs Age";
run;
```



Quadratic Regression Model: Steroid Level vs Age

The REG Procedure
Model: MODEL1
Dependent Variable: y

Number of Observations Read	27
Number of Observations Used	27

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	2	1046.26586	523.13293	52.63	<.0001
Error	24	238.54081	9.93920		
Corrected Total	26	1284.80667			

Root MSE	3.15265	R-Square	0.8143
Dependent Mean	17.64444	Adj R-Sq	0.7989
Coeff Var	17.86766		

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	95% Confidence Limits	
Intercept	1	-26.32541	5.88154	-4.48	0.0002	-38.46431	-14.18652
x	1	4.87357	0.77515	6.29	<.0001	3.27375	6.47340
x2	1	-0.11840	0.02347	-5.05	<.0001	-0.16684	-0.06996

- Regression seems to be fitting the data well, showing clear curvature.

$\widehat{\text{Steroid}} = -26.32541 + 4.87357\,\text{age} - 0.11840\,\text{age}^2$

- $R^2 = 0.8143$, 81.43% of variability in steroid level is explained by age and age^2.

b. Test whether or not there is a regression relation; use $\alpha = .01$. State the alternatives, decision rule, and conclusion. What is the P-value of the test? overall

$H_0: \beta_1 = \beta_2 = 0$
H_A: At least one β_1 or $\beta_2 \neq 0$

- $F = 52.63$
- $df1 = 2$
- $df2 = 24$
- p-value $< 0.0001 \ll 0.01 = \alpha$
- $\alpha = 0.01$

Reject H_0. Therefore, At least one β_1 or β_2 is a significant predictor. So there is a significant relationship between age and steroid level.

c. Obtain joint interval estimates for the mean steroid level of females aged 10, 15, and 20, respectively. Use the most efficient simultaneous estimation procedure and a 99 percent family confidence coefficient. Interpret your intervals.

```
/* Step 1: Create new input values */
data new;
    input x;
    x2 = x*x;
    y = .;
    datalines;
10
15
20
;
run;

/* Step 2: Combine original + new data */
data all;
    set steroid new;
run;

/* Step 3: Fit model & output predicted values + standard error */
proc reg data=all;
    model y = x x2 / alpha=0.01 clm; → Mean
    id x y x2;
    output out=ci_out p=pred stdp=stderr;
run;

/* Step 4: Compute Working-Hotelling 99% joint confidence intervals */
data joint_CI;
    set ci_out;
    if y = .; /* only new input values */
    fval = finv(0.99, 2, 24);  /* 2 predictors, n=27 → df = n - p - 1 = 24 */
    WH = sqrt(2 * fval);
    lowerWH = pred - WH * stderr;
    upperWH = pred + WH * stderr;
    keep x pred stderr lowerWH upperWH;
run;

proc print data=joint_CI noobs;
    title "Working-Hotelling 99% Joint Confidence Intervals for E(y)";
run;
```

Working-Hotelling (WH) best for estimating multiple means

$\boxed{\hat{y}_h \pm W \cdot SE(\hat{y}_h)}$

- $\hat{y}_h$ = predicted mean at age x_h
- $W = \sqrt{2 \cdot F_{2,\, n-3,\, 0.01}}$ → α
- $\mathrm{SE}(\hat{y}_h) = \sqrt{MSE \cdot \mathbf{h}_h}$, where $\mathbf{h}_h = \mathbf{x}_h'(X'X)^{-1}\mathbf{x}_h$
- X includes intercept, x, and x^2 → p

Using SAS output:

$W = \sqrt{2 \cdot F_{2,24,0.01}} = 3.4$

Working-Hotelling 99% Joint Confidence Intervals for E(y)

x	pred	stderr	lowerWH	upperWH
10	10.5702	0.92384	7.4747	13.6657
15	20.1379	0.89276	17.1465	23.1293
20	23.7856	0.85753	20.9122	26.6589

Intervals account for "Joint Estimation Uncertainty" over all 3 X-values, using WH.

- We are 99% confident that the true mean steroid level at:
  - Age 10 lies between 7.47 ~ 13.67
  - Age 15 lies between 17.15 ~ 23.13
  - Age 20 lies between 20.91 ~ 26.66

d. Predict the steroid levels of females aged 15 using a 99 percent prediction interval. Interpret your interval.

```
data new;
    input x;
    x2 = x*x;
    y = .;
    datalines;
15
;
run;

data all;
    set steroid new;
run;

                        Individual
                            ↑
proc reg data=all;
    model y = x x2 / alpha=0.01 cli;  /* cli gives prediction interval */
    id x;
run;
```

The REG Procedure
Model: MODEL1
Dependent Variable: y

Output Statistics

Obs	x	Dependent Variable	Predicted Value	Std Error Mean Predict	99% CL Predict		Residual
1	23	27.1	23.1325	1.0721	13.8188	32.4462	3.9675
2	19	22.1	23.5297	0.8804	14.3745	32.6848	-1.4297
3	25	21.9	21.5132	1.6078	11.6149	31.4114	0.3868
4	12	10.7	15.1077	0.7813	6.0232	24.1922	-4.4077
5	8	1.4	5.0855	1.4207	-4.5862	14.7572	-3.6855
6	12	18.8	15.1077	0.7813	6.0232	24.1922	3.6923
7	11	14.7	12.9574	0.8106	3.8528	22.0619	1.7426
8	8	5.7	5.0855	1.4207	-4.5862	14.7572	0.6145
9	17	18.6	22.3074	0.9223	13.1200	31.4948	-3.7074
10	18	20.4	23.0369	0.9072	13.8613	32.2125	-2.6369
11	9	9.2	7.9463	1.1301	-1.4209	17.3134	1.2537
12	21	23.4	23.8047	0.8635	14.6621	32.9473	-0.4047
13	10	10.5	10.5702	0.9238	1.3816	19.7588	-0.0702
14	25	19.7	21.5132	1.6078	11.6149	31.4114	-1.8132
15	9	11.8	7.9463	1.1301	-1.4209	17.3134	3.8537
16	17	24.6	22.3074	0.9223	13.1200	31.4948	2.2926
17	9	3.4	7.9463	1.1301	-1.4209	17.3134	-4.5463
18	23	22.8	23.1325	1.0721	13.8188	32.4462	-0.3325
19	13	21.1	17.0212	0.8062	7.9197	26.1228	4.0788
20	14	24.0	18.6980	0.8513	9.5644	27.8315	5.3020
21	16	21.8	21.3411	0.9180	12.1571	30.5251	0.4589
22	17	23.5	22.3074	0.9223	13.1200	31.4948	1.1926
23	21	19.4	23.8047	0.8635	14.6621	32.9473	-4.4047
24	24	25.6	22.4413	1.3010	12.9022	31.9803	3.1587
25	13	12.8	17.0212	0.8062	7.9197	26.1228	-4.2212
26	14	20.8	18.6980	0.8513	9.5644	27.8315	2.1020
27	18	20.6	23.0369	0.9072	13.8613	32.2125	-2.4369
28	15	.	20.1379	0.8928	10.9734	29.3024	.

- 15 Year-old female, predicted steroid level 20.14
- 99% PI for a randomly selected 15 Year-old: (10.97, 29.30)

e. Test whether the quadratic term can be dropped from the model; use $\alpha = .01$. State the alternatives, decision rule, and conclusion.

$H_0: \beta_2 = 0$ (Quadratic can be dropped)
$H_A: \beta_2 \neq 0$ (Quadratic can't be dropped)

From SAS output:

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	95% Confidence Limits	
Intercept	1	-26.32541	5.88154	-4.48	0.0002	-38.46431	-14.18652
x	1	4.87357	0.77515	6.29	<.0001	3.27375	6.47340
x2	1	-0.11840	0.02347	-5.05	<.0001	-0.16684	-0.06996

p-value: $< 0.0001 \lll 0.01 = \alpha$

Reject Ho. Therefore, there is strong evidence that the X^2 Quadratic term cannot be dropped and is significant to the model.

f. Express the fitted regression function obtained in part (a) in terms of the original variable X.

$\hat{Y} = -26.32541 + 4.87357X - 0.11840X^2 + \varepsilon_i$

---

6. KNNL Problem 8.15

8.15. Refer to **Copier maintenance** Problem 1.20. The users of the copiers are either training institutions that use a small model, or business firms that use a large, commercial model. An analyst at Tri-City wishes to fit a regression model including both number of copiers serviced (X_1) and type of copier (X_2) as predictor variables and estimate the effect of copier model (S—small, L—large) on number of minutes spent on the service call. Records show that the models serviced in the 45 calls were:

i:	1	2	3	...	43	44	45
X_{i2}:	S	L	L	...	L	L	L

Assume that regression model (8.33) is appropriate, and let $X_2 = 1$ if small model and 0 if large, commercial model.

a. Explain the meaning of all regression coefficients in the model.

$y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$

- y: Minutes spent on service call
- X_1: Number of copiers serviced
- X_2: Copier model (1 = Small, 0 = large)

β_0: Expected Service Time (in minutes) for a large commercial model, ($X_2 = 0$), when 0 copiers are serviced ($X_1 = 0$)
- Baseline

β_1: Increase in Service Time (in minutes) for each additional copiers serviced, adjusting for copier time.

β_2: Difference in Service Time (in minutes) between small and large models, controlling for the number of copiers serviced.

b. Fit the regression model and state the estimated regression function.

```
proc reg data=copier;
    model y = x1 x2 / clb;
    title "Regression Model: Service Time vs Number of Copiers and Model Type";
run;
quit;
```

Regression Model: Service Time vs Number of Copiers and Model Type

The REG Procedure
Model: MODEL1
Dependent Variable: y

Number of Observations Read	45
Number of Observations Used	45

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	2	77001	38501	479.04	<.0001
Error	42	3375.53856	80.36997		
Corrected Total	44	80377			

Root MSE	8.96493	R-Square	0.9580
Dependent Mean	76.26667	Adj R-Sq	0.9560
Coeff Var	11.75472		

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	95% Confidence Limits	
Intercept	1	-1.37162	3.03082	-0.45	0.6532	-7.48806	4.74481
x1	1	15.05150	0.48641	30.94	<.0001	14.06989	16.03312
x2	1	1.99232	2.79494	0.71	0.4799	-3.64809	7.63273

$\hat{Y} = -1.37162 + 15.05150X_1 + 1.99232X_2$

c. Estimate the effect of copier model on mean service time with a 95 percent confidence interval. Interpret your interval estimate.

95% CI: (-3.64809, 7.63273)

- We are 95% confident that [β_2: Difference in Service Time (in minutes) between small and large models, controlling for the number of copiers serviced.] lies between -3.65 and 7.63 minutes.
- Since this interval contains 0, we fail to reject H_0, therefore cannot conclude copier model predictor is significant.

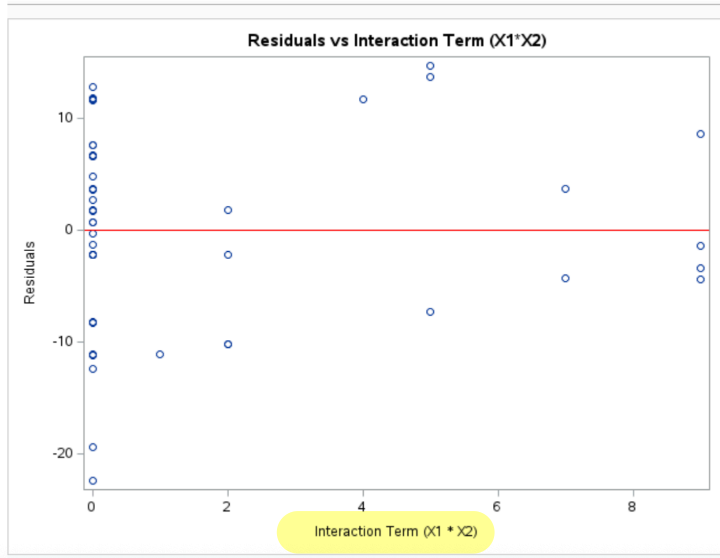
d. Why would the analyst wish to include X_1, number of copiers, in the regression model when interest is in estimating the effect of type of copier model on service time?

- Include X_1 as a possible confounder; may also affect total service time regardless of model type.
- X_1 controlles for number of copiers serviced, which directly affects service time, making sure model type would not be confounded by differences in workload.

e. Obtain the residuals and plot them against $X_1 X_2$. Is there any indication that an interaction term in the regression model would be helpful?

```
/* Fit model and output residuals */
proc reg data=copier;
    model y = x1 x2;
    output out=residout r=resid;
run;
quit;

/* Plot residuals vs interaction term */
proc sgplot data=residout;
    scatter x=x1x2 y=resid;
    refline 0 / axis=y lineattrs=(color=red);
    xaxis label="Interaction Term (X1 * X2)";
    yaxis label="Residuals";
    title "Residuals vs Interaction Term (X1*X2)";
run;
```



- Noticable Spread and variability at certain interaction values, suggesting the relationship between X_1 and Y might depend on X_2.
- Adding interaction would improve fit.

STAT525 HOMEWORK#5 Solutions

1. Use the **patient satisfaction** data described in KNNL Problem 6.15.

   a Compute the pairwise correlations between the X's and between each X and Y. Which X variable appears to be the best individual predictor?

   There do not appear to be any really strong correlations among the explanatory variables (all < 0.70). Age ($x1$) appears to be the best individual predictor since it has the largest correlation ($r = -0.79$).

```
          Pearson Correlation Coefficients, N = 46
                    Prob > |r| under H0: Rho=0
                x1           x2           x3            y
x1         1.00000      0.56795      0.56968     -0.78676
                         <.0001       <.0001       <.0001
x2         0.56795      1.00000      0.67053     -0.60294
            <.0001                    <.0001       <.0001
x3         0.56968      0.67053      1.00000     -0.64459
            <.0001       <.0001                    <.0001
y         -0.78676     -0.60294     -0.64459      1.00000
            <.0001       <.0001       <.0001
```

   b Run the linear regression with age, severity of illness and anxiety level as the explanatory variables and satisfaction as the response variable. Summarize the regression results.

   The fitted regression line is $\widehat{Y} = 158.49 - 1.14X_1 - 0.44X_2 - 13.47X_3$ where X_1 is age, X_2 is severity level, and X_3 is anxiety level. The coefficient of determination is $R^2 = 0.6822$ and the P-value of the F test is $< .0001$ suggesting that this set of explanatory variables is helpful in explaining satisfaction. The test is testing the hypotheses

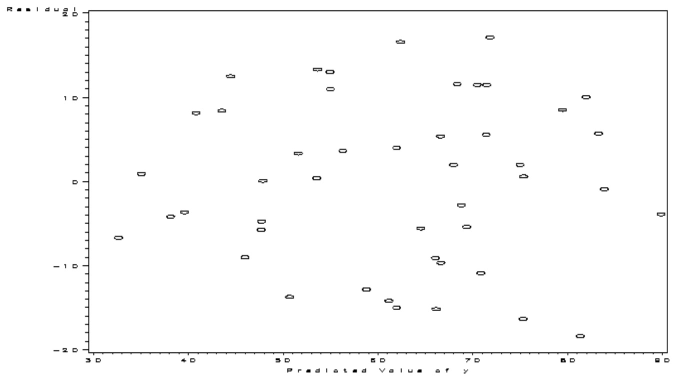
$$H_0 : \beta_1 = \beta_2 = \beta_3 = 0$$
$$H_a : \text{at least one } \beta_i \neq 0$$

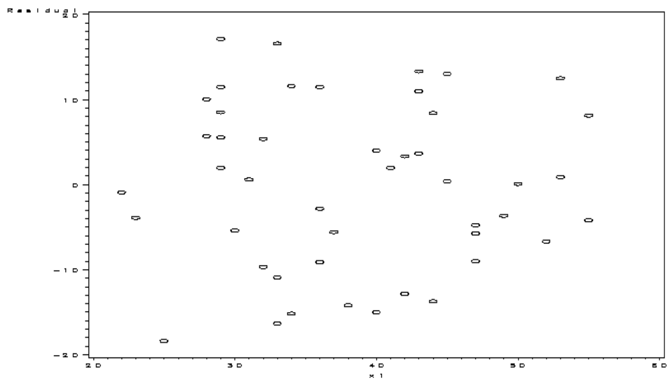
   The test statistic is 30.05 and the degrees of freedom are 3 and 42.

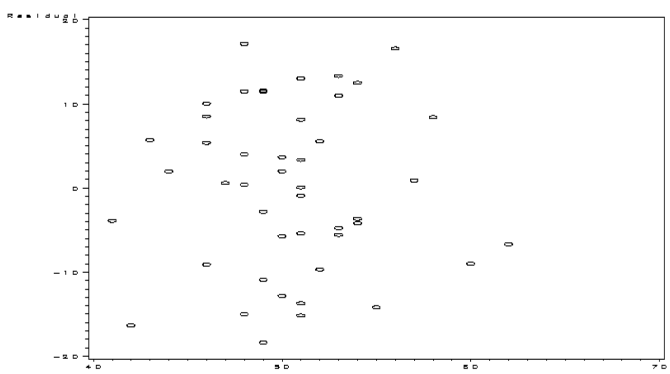
   Only one individual parameter t-tests is significant (Age) suggesting that perhaps not all variables are needed in predicting satisfaction. All of the coefficients are negative suggests an increase in any one of them results in an decrease in satisfaction.

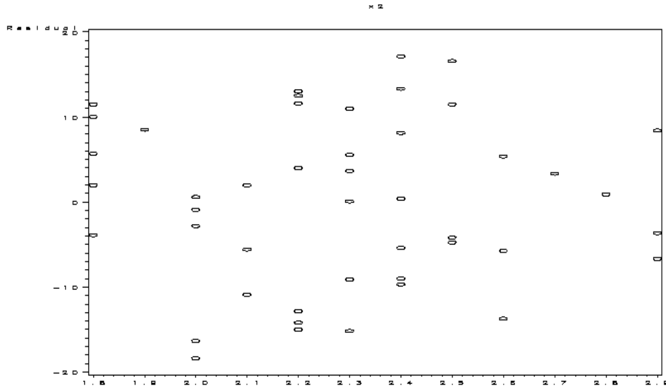
   c Plot the residuals versus the predicted satisfaction and each of the explanatory variables. Are there any unusual patterns?

   I do not see any unusual patterns in any of the residual plots.


Residual
Predicted Value of y

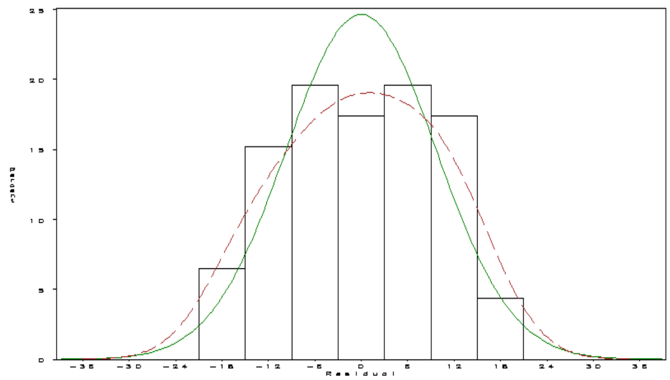

Residual
x1

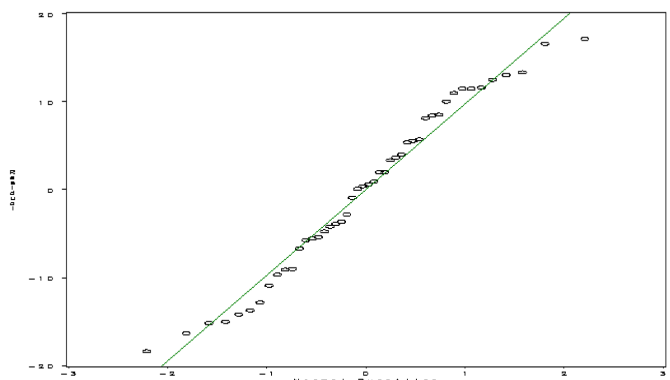

Residual
x2


Residual
x3

d Examine the assumption of normality for the residuals using a qqplot or histogram. State your conclusions.

The data appears to be reasonably normal. No real deviations can be found in these plots.





e Predict the satisfaction for a 55 year old patient with illness severity 50 and anxiety level 2.8. Provide a 95% prediction interval with your prediction.

The following is from SAS using the CLI option.

```
        Dependent    Predicted       Std Error
Obs      Variable        Value    Mean Predict      95% CL Predict
47              .      35.8859          4.6332   13.5381       58.2338
```

2. KNNL Problem 7.5

This can be done using Type I sum of squares. The following output summarizes the various extra sums of squares. As far as the test in part b, this is the same as the t-test since the X_3 variable is fitted last. In this case, we would not reject and conclude that this variable could be dropped given the other two are already in the model. The test is testing the hypotheses

$$H_0 : E(Y) = \beta_0 + \beta_1 X_1 + \beta_2 X_2$$
$$H_a : E(Y) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3$$

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	9120.46367	3040.15456	30.05	<.0001
Error	42	4248.84068	101.16287		
Corrected Total	45	13369			

Root MSE	10.05798	R-Square	0.6822
Dependent Mean	61.56522	Adj R-Sq	0.6595
Coeff Var	16.33711		

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Type I SS
Intercept	1	158.49125	18.12589	8.74	<.0001	174353
x2	1	-0.44200	0.49197	-0.90	0.3741	4860.26000
x1	1	-1.14161	0.21480	-5.31	<.0001	3896.04414
x3	1	-13.47016	7.09966	-1.90	0.0647	364.15952

3. KNNL Problem 7.6

   This can be done using the TEST option. The output below summarizes the test. Using $\alpha = .025$, we would reject the null hypothesis and assume that we cannot throw both the variables out of the model.

Source	DF	Mean Square	F Value	Pr > F
Numerator	2	422.53741	4.18	0.0222
Denominator	42	101.16287		

4. KNNL Problem 7.14

   These can be obtained using the `pcorr1` `pcorr2` options (interpretations of type I and type II are similar to `ss1` and `ss2`, respectively).

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Squared Partial Corr Type I	Squared Partial Corr Type II
Intercept	1	158.49125	18.12589	8.74	<.0001	.	.
x1	1	-1.14161	0.21480	-5.31	<.0001	0.61898	0.40211
x2	1	-0.44200	0.49197	-0.90	0.3741	0.09441	0.01886
x3	1	-13.47016	7.09966	-1.90	0.0647	0.07894	0.07894

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Squared Partial Corr Type I	Squared Partial Corr Type II
Intercept	1	158.49125	18.12589	8.74	<.0001	.	.
x2	1	-0.44200	0.49197	-0.90	0.3741	0.36354	0.01886
x1	1	-1.14161	0.21480	-5.31	<.0001	0.45787	0.40211
x3	1	-13.47016	7.09966	-1.90	0.0647	0.07894	0.07894

The above outputs report these values:

- $r^2_{Y1} = 0.61898$
- $r^2_{Y1|2} = 0.45787$
- $r^2_{Y1|23} = 0.40211$
- $r^2_{Y2} = 0.36354$
- $r^2_{Y2|1} = 0.09441$
- $r^2_{Y2|13} = 0.01886$

Based on these on can see that $x1$ is strongly associated with y and does not change all that much given other variables in the model. Variable $x2$, however, changes dramatically depending on what is in the model.

5. KNNL Problem 7.18

It really does not matter whether you use the standardization transformation of the correlation transformation in this case. Thus, the stb option will compute the coefficients and give you the coefficients in the original units too.

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	9120.46367	3040.15456	30.05	<.0001
Error	42	4248.84068	101.16287		
Corrected Total	45	13369			

Root MSE	10.05798	R-Square	0.6822
Dependent Mean	61.56522	Adj R-Sq	0.6595
Coeff Var	16.33711		

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Standardized Estimate
Intercept	1	158.49125	18.12589	8.74	<.0001	0

```
x1          1   -1.14161   0.21480    -5.31   <.0001  -0.59067
x2          1   -0.44200   0.49197    -0.90   0.3741  -0.11061
x3          1  -13.47016   7.09966    -1.90   0.0647  -0.23393
```

The coefficients of determination among all pairs of predictor variables is simply the correlation squared (see the correlation coefficients in Q1.a). Since this is a linear transformation, the correlation between the untransformed and transformed variables are the same. This means that a similar type of interpretation can be made regarding each coefficient. The difference is now the change is in terms of standard deviation units.

6. Derive equation 7.56 (HINT: Recall that b_1 can be obtained by regressing the residuals of $Y|X_2$ vs $X_1|X_2$)

   The first relationship below uses the fact that b_1' can be obtained by regressing the residuals of $Y|X_2$ vs $X_1|X_2$. The second plugs in the fitted values using b and $b^\star$ for the slope estimates. The remainder uses the relationship between the Pearson's correlation coefficient and the slope.

$$\begin{aligned} b_1 &= \frac{\sum(X_{i1}-\hat{X}_{i1})(Y_i-\hat{Y}_i)}{\sum(X_{i1}-\hat{X}_{i1})^2} \\ &= \frac{\sum(X_{i1}-\overline{X}_1-b(X_{i2}-\overline{X}_2))(Y_i-\overline{Y}-b^\star(X_{i2}-\overline{X}_2))}{\sum(X_{i1}-\overline{X}_1-b(X_{i2}-\overline{X}_2))^2} \\ &= \frac{\sum(X_{i1}-\overline{X}_1)(Y_i-\overline{Y})-2b\sum(X_{i2}-\overline{X}_2)(Y_i-\overline{Y})+bb^\star\sum(X_{i2}-\overline{X})^2}{\sum(X_{i1}-\overline{X}_1)^2-b^2\sum(X_{i2}-\overline{X}_2)^2} \\ &= \frac{\sum(X_{i1}-\overline{X}_1)(Y_i-\overline{Y})-2r_{12}r_{Y2}(n-1)s_1s_Y+r_{12}r_{Y2}(n-1)s_1s_y}{\sum(X_{i1}-\overline{X}_1)^2-b^2\sum(X_{i2}-\overline{X}_2)^2} \\ &= \frac{b_1'-\frac{r_{12}r_{Y2}s_Y}{s_1}}{1-r_{12}^2} \end{aligned}$$

   You can also start from equations in 6.77 (page 241): eliminate b_0 by replacing Y_i with $Y_i-\bar{Y}$, X_{i1} with $X_{i1}-\bar{X}_{\cdot 1}$, $X_{i2}-\bar{X}_{\cdot 2}$.

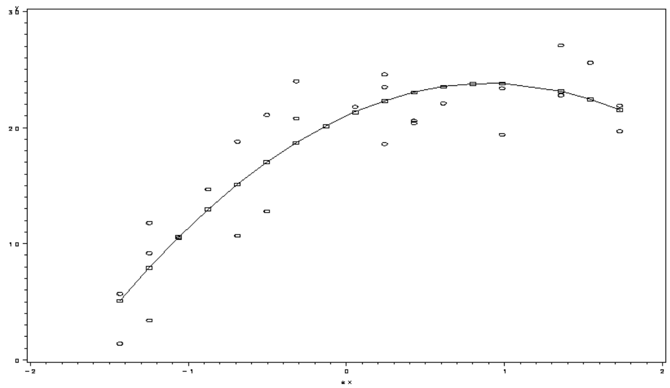
7. KNNL Problem 8.6

   Based on the plot below, a quadratic relationship appears to be a good fit.

   The regression output is also shown. The overall model is significant. This is testing $H_0:\beta_1=\beta_2=0$ versus the alternative that at least one of them is not equal to zero. Since the P-value is less than .01, we reject the null. Looking at the t-tests, both regression coefficificents are significantly different from zero. The R^2 here is equal to 81.43%.

```
                          Analysis of Variance

                                   Sum of           Mean
Source                    DF      Squares         Square    F Value    Pr > F
```



Model	2	1046.26586	523.13293	52.63	<.0001
Error	24	238.54081	9.93920		
Corrected Total	26	1284.80667			

Root MSE	3.15265	R-Square	0.8143
Dependent Mean	17.64444	Adj R-Sq	0.7989
Coeff Var	17.86766		

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	21.00498	0.91255	23.02	<.0001
sx	1	6.21404	0.62556	9.93	<.0001
sx2	1	-3.42261	0.67840	-5.05	<.0001

The Bonferroni approach was used to obtain simultaneous CI's. For an overall 99% level, we'd use 99.67% level for each of the three CI's.

x	Predicted Value	Std Error Mean Predict	99.66667% CL Mean	
10	10.5702	0.9238	7.5600	13.5804
15	20.1379	0.8928	17.2290	23.0469
20	23.7856	0.8575	20.9914	26.5798

The prediction interval for age=15 is $(10.9734, 29.3024)$. This means that with 99% confidence, we believe the observed response will be between these two numbers.

The individual t-test can be used to assess whether the quadratic term can be dropped. In this case, it cannot (see t-test result above). Note that $sx = \frac{X-15.7778}{5.5006}$. The model, rewritten in terms of X is $E(Y) = -26.32541 + 4.87357X - 0.1184X^2$.

8. KNNL Problem 8.15

It is assumed that there is the same rate of change between the number of copiers serviced and the number of minutes spent on the service call (slope is the same). The parameter on X_2 represents the difference in slopes between the small and large copier models. The regression model is shown below

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	2	76966	38483	473.94	<.0001
Error	42	3410.32825	81.19829		
Corrected Total	44	80377			

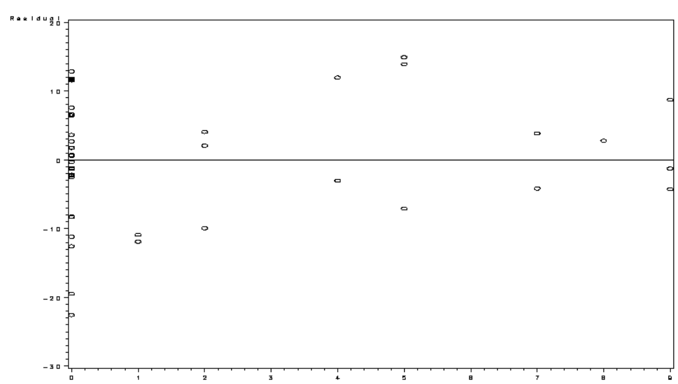
Root MSE	9.01101	R-Square	0.9576
Dependent Mean	76.26667	Adj R-Sq	0.9556
Coeff Var	11.81513		

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	95% Confidence Limits	
Intercept	1	-0.92247	3.09969	-0.30	0.7675	-7.17789	5.33294
x1	1	15.04614	0.49000	30.71	<.0001	14.05728	16.03500
x2	1	0.75872	2.77986	0.27	0.7862	-4.85125	6.36870

The stated model is $E(Y) = -0.92247 + 15.05X_1 + .75872X_2$. This implies that for the same number of copiers, the service call lasts .75872 minutes longer for the small compared to the large copiers. It should also be noted that as the number of copiers serviced increases, the longer the number of minutes spent on the service call. By including this in the model, we reduce the unexplained variation and thus make our comparison between large and small copiers more precise. It also allows us to make the large versus small comparison at the same level of copiers serviced.

The plot below shows the residuals versus X_1X_2. There does not appear to be any linear trend and the residuals are scattered randomly. I would not expect this to be a helpful predictor.



```
/* SAS Codes for Q1 */
```

```
data hw5q1;
    infile 'd:\nobackup\tmp\ch06pr15.txt';
    input y x1 x2 x3;

proc corr data=hw5q1;
    var x1 x2 x3 y;
run;

data newobs;
    x1=55; x2=50; x3=2.8;
    output;
run;

data thw5q1;
    set hw5q1 newobs;
run;

proc reg data=thw5q1;
    model y=x1 x2 x3/cli;
    output out=q1out r=resid p=pred;
run;

symbol v=circle i=none c=black;
proc gplot data=q1out;
    plot resid*(pred x1 x2 x3);
run;

proc univariate data=q1out;
    histogram resid / normal kernel (L=2);
    qqplot resid / normal (L=1 mu=est sigma=est);
run; quit;

/* SAS Codes for Q2 & Q3 */
proc reg data=hw5q1;
    model y=x2 x1 x3/ss1;
    Q3: test x2, x3;
run; quit;

/* SAS Codes for Q4 */
proc reg data=hw5q1;
    model y=x1 x2 x3/pcorr1 pcorr2;
run;

proc reg data=hw5q1;
    model y=x2 x1 x3/pcorr1 pcorr2;
run; quit;
```

```
/* SAS Codes for Q5 */
proc reg data=hw5q1;
    model y=x1 x2 x3/stb;
run; quit;

/* SAS Codes for Q7 */
goptions colors=(none);
data hw5q7;
 infile 'D:\nobackup\tmp\ch08pr06.txt';
 input y x;
 sx = x;

data newobs;
 input x y;
 sx = x;
 cards;
 10 .
 15 .
 20 .
;

data thw5q7;
    set hw5q7 newobs;

proc standard data=thw5q7 out=nhw5q7 mean=0 std=1;
    var sx;

data nhw5q7;
    set nhw5q7;
    sx2 = sx*sx;

proc reg data=nhw5q7;
    model y=sx sx2 / clm alpha=.00333333333333333333;
    id x y;
    output out=q7out p=pred;
run;

symbol1 v=circle i=none;
symbol2 v=square i=join;
proc sort data=q7out; by sx;
proc gplot data=q7out;
    plot y*sx pred*sx / overlay;
run;

proc reg data=nhw5q7;
    model y=sx sx2/ cli alpha=.01;
```

```
    id x;
    Q7e: test sx2;
run;

proc univariate data=hw5q7;
    var x;
    output out=xstat mean=mean std=std;
run; quit;

/* SAS Codes for Q8 */
goptions colors=(none);
data hw5q8a;
    infile 'D:\nobackup\tmp\ch01pr20.txt';
    input y x1;

data hw5q8;
    set hw5q8a;
    infile 'D:\nobackup\tmp\ch08pr15.txt';
    input x2;
    x12 = x1*x2;

proc reg data=hw5q8;
    model y=x1 x2 / clb alpha=.05;
    output out=q8out r=resid;
run;

proc sort data=q8out; by x12;
symbol1 v=circle i=none;
proc gplot data=q8out;
    plot resid*x12/vref=0;
run;quit;
```